

SECRETARY ISLAND, New Zealand

WMO: 938000

Lat: 45.221S

Lon: 166.885E

Elev: 19

StdP: 101.10

Time Zone: 12.00 (NZL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 7	(b) 3.3	(c) 4.2	(d) -2.2	(e) 3.2	(f) 6.4	(g) -0.8	(h) 3.5	(i) 6.5	(j) 14.8	(k) 11.5	(l) 13.6	(m) 11.1	(n) 2.8	(o) 150	(p) 0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	5.6	21.2	16.2	20.1	15.9	19.1	15.6	17.8	19.5	17.0	18.9	16.3	18.1	4.0	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	18.6	16.2	11.6	17.9	15.4	11.0	17.3	50.4	19.5	48.0	19.0	45.7	18.3	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 14.2	(b) 12.8	(c) 11.4	(e) 2.0	(f) 23.9	(g) 0.7	(h) 1.5	(i) 1.5	(j) 24.9	(k) 1.0	(l) 25.8	(m) 0.6	(n) 26.6	(o) 0.1	(p) 27.7		
(a) 4	(b) 5	(c) 6	(e) 0.5	(f) 18.4	(g) 0.6	(h) 1.2	(i) 0.1	(j) 19.3	(k) -0.3	(l) 20.0	(m) -0.6	(n) 20.7	(o) -1.1	(p) 21.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.8	14.8	15.1	14.3	12.8	11.2	9.6	9.1	9.5	10.1	10.6	11.6	13.4
	DBStd	2.85	2.22	2.08	2.08	1.93	2.10	1.83	1.78	1.69	1.78	1.83	2.00	2.15
	HDD10.0	152	0	0	1	2	11	29	38	29	20	14	7	1
	HDD18.3	2381	112	92	127	166	219	262	286	274	247	241	203	152
	CDD10.0	817	148	143	133	86	50	17	11	13	23	32	54	108
	CDD18.3	4	2	1	1	0	0	0	0	0	0	0	0	0
	CDH23.3	4	2	1	1	0	0	0	0	0	0	0	0	1
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	4.5	4.7	4.6	4.5	4.3	4.3	3.8	4.2	4.2	4.8	4.8	4.8	4.8
Precipitation	PrecAvg	3808	359	300	324	307	364	282	252	259	297	358	349	362
	PrecMax	4852	626	639	771	714	674	460	386	445	429	611	694	589
	PrecMin	2940	76	129	103	122	151	112	105	115	75	198	134	144
	PrecStd	416	130	117	146	129	112	91	73	76	95	114	134	113
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.1	22.6	21.5	19.2	17.6	14.9	14.6	15.3	16.5	18.1	19.8	21.9
		MCWB	17.0	16.7	15.6	13.7	13.5	11.6	10.3	10.2	11.7	12.4	14.3	16.1
	2%	DB	21.2	20.9	19.8	17.7	16.0	13.9	13.3	13.7	15.0	16.0	17.6	19.9
		MCWB	16.5	16.3	15.8	13.8	13.2	11.4	10.2	10.3	11.1	11.9	13.1	15.4
	5%	DB	19.9	19.7	18.7	16.7	15.1	13.2	12.6	12.9	13.8	14.6	16.0	18.3
		MCWB	16.1	16.1	15.7	13.6	12.8	11.2	10.1	10.1	10.6	11.4	12.6	14.9
	10%	DB	18.6	18.7	17.8	15.8	14.3	12.5	11.8	12.1	12.9	13.6	14.8	17.1
		MCWB	15.4	15.9	15.2	13.4	12.3	10.7	9.8	9.8	10.4	11.2	12.3	14.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.0	18.7	18.5	15.9	15.4	13.4	12.5	12.2	13.1	13.9	15.6	17.4
		MCDB	20.6	20.3	19.2	17.0	16.2	13.9	13.3	13.1	14.9	15.7	17.7	19.9
	2%	WB	17.7	17.7	17.4	15.1	14.3	12.5	11.6	11.4	12.1	13.0	14.2	16.4
		MCDB	19.8	19.4	18.6	16.3	15.2	13.2	12.5	12.6	13.4	14.2	15.9	18.3
	5%	WB	16.7	17.0	16.4	14.5	13.5	11.7	10.9	10.8	11.6	12.4	13.5	15.6
		MCDB	18.8	18.8	17.9	15.8	14.5	12.7	11.9	12.2	12.9	13.7	15.2	17.5
	10%	WB	15.9	16.2	15.6	13.8	12.7	11.0	10.2	10.2	11.0	11.7	12.8	14.8
		MCDB	17.8	18.1	17.3	15.3	13.9	12.1	11.4	11.7	12.5	13.2	14.4	16.9
Mean Daily Temperature Range	5% DB	MDBR	5.6	5.6	5.3	5.1	4.5	4.5	4.6	4.9	4.9	5.0	5.3	5.6
		MCDBR	7.9	7.5	7.1	6.3	5.2	5.2	5.4	5.9	6.4	6.9	7.5	7.6
		MCWBR	4.9	4.7	4.5	4.3	3.8	3.9	3.9	4.0	4.4	4.5	4.9	4.8
	5% WB	MCDBR	6.0	5.8	4.9	4.5	4.0	3.9	3.5	4.5	4.9	4.9	5.4	6.0
		MCWBR	4.2	4.2	3.8	3.9	3.6	3.7	3.3	3.7	3.9	3.8	4.1	4.1
Clear-Sky Solar Irradiance	taub	0.351	0.340	0.325	0.317	0.311	0.305	0.299	0.312	0.348	0.342	0.339	0.336	
	taud	2.528	2.575	2.625	2.628	2.607	2.577	2.608	2.592	2.470	2.506	2.532	2.574	
	Ebn at Noon	970	952	920	851	773	733	779	842	875	939	976	992	
	Edh at Noon	107	96	83	69	59	54	57	70	95	102	106	103	
All-Sky Solar Radiation	RadAvg	5.49	4.75	3.52	2.31	1.35	1.07	1.27	1.99	2.96	4.15	5.21	5.74	
	RadStd	0.47	0.36	0.26	0.20	0.12	0.12	0.13	0.12	0.27	0.32	0.53	0.42	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.31	+0.32	N/A	N/A	N/A	N/A	-30	-125	+84
(47) Regional (3 neighbors)		N/A	N/A	+0.77	N/A	N/A	N/A	-47	-112	N/A

Nomenclature: See separate page